A-Team Tutorials
Level 2 with Tycho Tracker

Targets and Priorities

The objective at Level 2 is to identify targets that are of scientific interest.

In general, the desirability of observations comes from having an uncertain orbit, and no recent observations. Desirability is much increased if the object has a degree of "risk" associated – a recently discovered Virtual Impactor (VI) is an object for which a possible future Earth impact has not been ruled out and would therefore carry a high priority.

Different organisations assign priority in different ways and there are multiple sources of information that are sometimes confusing.

NEOFixer

The main source of NEO priority information is now NEOFixer. This site is used by professional observatories as well as amateurs. You will need to register and set up your profile to show targets for the Slooh telescopes.

Once you have an account, select your profile (Username) and add the telescopes E62, G40 and W88 using the Telescopes tab and entering the three Observatory codes. Eventually you will get a confirmation from NEOFixer.

On the Targets page you can view a list of objects visible from a specified observatory. On the Activity page you can view targets for any of your "authorised" telescopes.

There are lots of options to sort and filter the lists. Read the FAQ.

When you select an object, you can obtain various pieces of information about it and generate ephemeris.

MPC "Observable" Lists

There are several lists published by the MPC but most useful is the customisable list that allows the user to set limits (e.g., for magnitude) and to select different types of objects. The page can be found at the Observable Object List Customizer:

http://www.minorplanetcenter.net/iau/lists/Customize.html

Some of the lists are quite long and, just because an object appears on a list, it does not necessarily mean the MPC urgently requires observations. There may well be observations "in the pipeline" that the MPC has not yet processed (and in practice I sometimes find objects in these lists that should not be there).

The lists are structured in three parts, and you should read the page headings carefully to understand what is included in each part. The displayed data varies from list to list but there are some key things to focus on:

- "Currently" shows low-resolution RA and Dec, V-magnitude, Elongation and Motion. Elongation refers to angular distance from the Sun. Check that the V-magnitude is within your capabilities and that Motion is less that about 5.0″ (arcseconds per minute). For faster objects, (up to around 20″ /m) you may be able to use the Multi-Luminance 20s recipe if the object is bright enough.
- "Last obs" tells you when it was last seen. If it has not been seen for a while, observations may be needed and there are various coded signals next to observation dates (read the introductions at the top of the list to find what they mean).
- Check the column "U" (uncertainty of the orbit). Objects of zero uncertainty may not be of much interest, but objects of very high uncertainty should be avoided. An uncertainty of 9 means "this could be anywhere" so don't waste missions looking for it! Objects with 1<u<6 may need observations and should be relatively easy to find.

MPC NEO Observation Planning Aid

The MPC provides a NEA Observation Planning Aid. It is a bit complicated but you can try it.

International Asteroid Warning Network

This page lists objects that will pass within one lunar distance. It provides direct links to the MPC pages related to the object.

Goldstone Observation schedule

The Goldstone radar station targets are listed for the upcoming weeks and there maybe something of interest there.

Minor Planet Mailing List (MPML)

On groups.io there is the Minor Planet Mailing List that you can subscribe to. There is always lots of chatter about current targets. Most of it is from professionals and involves objects too faint or fast for us, but sometimes there is an interesting target.

A-Team.space

A targets list can be found at the A-Team.space website. This is a simplified list based on data from NEOFixer. Use the filter and observatory buttons to list objects of interest. You can select them and generate ephemeris from the MPC.

Visibility and Missions

At Level 2 you are trying to capture an object of scientific value. It may well be fainter than objects observed previously and requires more careful planning.

Ephemerides

There are several different ways of obtaining ephemerides (and the data needed to complete the checklist discussed below). You may wish to try them and decide on your favourite:

Minor Planet Centre Ephemeris Page

http://www.minorplanetcenter.net/iau/MPEph/MPEph.html

This is the one I recommended in Level 1 and remains my favourite. If you are OK with this one, don't bother about the others.

NEODyS (for NEOs) and AstDyS (Other asteroids)

These are sites maintained by ESA. They contain more information but are quite complex to use.

- NEODyS-2
- ASTDyS-2

Go to Objects, and search to find your target and you will see links to obtain all the information you might need.

JPL Horizons

HORIZONS Web-Interface (nasa.gov) is probably the most accurate for longer term predictions of highly perturbed objects but not always fully up to date on the latest discoveries.

Lowell Observatory ASTEPH

Lowell astorb DB. Beware this may not have the latest orbits for recent discoveries and close approach objects.

Planetarium Software.

Any one of the many PC and on-line packages may supply coordinates and other data. Make sure you understand how current the orbit data is. Close-approach NEOs will not be shown correctly if the software is using old data.

Find_Orb

You can download the currently published observations from the MPC Explorer and use Find_Orb to generate the ephemeris. Use of Find_Orb is described in a separate tutorial.

Tycho Tracker

If you are using Tycho Tracker, it has a Session Planning module you can try.

Visibility

You will need to obtain coordinates and check several factors:

Magnitude:

It is very difficult to give an exact guide as to how faint you will be able to go. On a good night (good seeing, no Moon, no hazy clouds etc) you may get reasonable results down to V=17 or 18. It is probably best to stay at least a magnitude brighter to practice missions and measurements.

If you use the Multi-luminance recipe for your missions you will be able to “stack” multiple images for each observation and achieve 1 or 2 magnitudes fainter. On poor visibility nights the limit is 1 – 2 magnitudes brighter than on good nights.

Altitude

Go for objects that are well above 30 degrees from the horizon. Higher altitude will give access to fainter objects. (I would normally like my objects above 40 degrees). Check which of the Slooh observatories will give you the best view.

Motion

Fast objects will leave faint trails and will be difficult to measure: Ideally the object should not move more than 3 or 4 pixels during the exposure. Work out what is the maximum speed for the length of integration you are using. In practice up to 5 or 6 arcseconds/min is manageable with 50-second exposures, maybe a bit more with practice and good clear images. For bright objects you can use 20-second exposures and get good results for objects up to 20”/m provided they are bright enough to be measured accurately after such short exposures.

Moon

Consider its impact on visibility. Remember Slooh will not allow a reservation if it thinks the Moon is too close and too bright. T1 and A1 are more sensitive to Moonlight coming in from the side than the other telescopes. I would normally like the Moon 60 degrees away if it is more than 50% full. The Moon is less of a problem when it is close to the horizon than when it is high up. A bright Moon in the sky can easily reduce your limit by two magnitudes.

Uncertainty

Most objects will have only a few arcseconds of uncertainty and that is of little concern. Objects that have not been observed for a long time may have much larger uncertainty and risk not being found in the FOV of the telescope. The MPC ephemeris usually tells you the uncertainty in arcseconds or provides an “uncertainty map”. Beware objects that have not been observed for a long time and have no uncertainty data – they are probably “lost”, and the ephemeris is meaningless.

Galactic Latitude

The Galactic equator is the plane of the disk of the Milky Way galaxy and therefore is the direction of the mass of Milky Way stars. Targets close to the Galactic Equator may get lost in the glare of stars or be too close to a star for accurate measurement. It is best to look above or below ~20 degrees of Galactic Latitude.

Missions

Check what slots are available for the coming night. Check the weather forecast for Chile, Canary Isles and Australia.

If you intend to make an MPC report, you should try for 3 good measurements spread out over sufficient time that the object has moved significantly. The missions should be about 10 minutes apart (or more for a very slow object). Clearly there will always be a compromise between the factors listed in the ephemeris and the availability of mission slots.

If you have a Slooh double-membership, schedule three pairs of consecutive missions.

Multi Luminance

Missions should be set up using the Multi-Luminance recipe. This will yield multiple images for each mission, and we can use “Stack and Track”. That means multiple images are averaged to get a better Signal-to-Noise ratio.

Select the mission slots you want and put them on 1 hour hold. Then go back and set up the coordinates. That way nobody can jump in and take one of the slots you wanted.

The following is specific to Astrometrica.

Identify and Measure

The main difference between Level 1 and 2 is that we are looking to create useful observations of a Near Earth object (or other interesting target) and report them to the MPC.

At Level 2 we are probably dealing with a fainter target. The missions should be “Mult Luminance” and hopefully you will have collected three groups of images with several images in each group.

Prepare Images

Previously, we stepped through Calibrate, Plate Solve and Alignment. Now we should set up Action/Express Mode to do all the steps quickly. The first time you use Express Mode, check the settings are OK in each category. Subsequently just start Express Mode and everything happens quickly.

Make sure “clean-up intermediate directories” is checked and “Auto load output files into Image Manage” is checked in Settings/Image Directories. That way your calibrated, plate-solved and aligned images will be places in the Image Manager list automatically.

If not already open, Action/View Images.

Identify Target

There are several ways to identify your target.

Known Objects

This is the approach used in Level 1 and will often be sufficient. However, close-approach NEOs or other recently discovered objects may not have good, published orbits in MPCORB so the object is not shown in the right position or anywhere in the field of view.

JPL Ephemeris

NASA JPL generally has a pretty good orbit for everything but the most recent discoveries.

- Image Manager/Ephemeris/Attach from JPL Horizons.
- Enter the object name and Check Object to download the orbit data.
- JPL Horizon Interface/Ephemeris/Attach to dataset will put the object's predicted position in each line of the Image Manager.
- Image Viewer/Create Stack – Ephemeris will stack the images with the objects position in or near the centre.
- Double-click the object to centre it.
- Image Viewer/Create/Track from current position will produce a track from which you can Verify and Create Observations as before.

Find_Orb Ephemeris

If there are any existing observations (including your own), a new orbit can be calculated and used to generate Ephemeris.

- Tools/Download Observations/ downloads all observations.
- Text form-observations/Compute Orbit will produce orbital parameters using the version of Find_Orb selected in Settings.
- Orbit/Ephemeris/Attach to dataset will put the object's predicted position in each line of the Image Manager.
- Image Viewer/Create Stack – Ephemeris will stack the images with the objects position near the centre.
- Double click the object to centre it.
- Image Viewer/Create/Track from current position will produce a track from which you can Verify and Create Observations as before.

It is now worth considering installing the local Find_Orb application as described in the Find_Orb tutorial. It gives an extra degree of control of the orbit generation process and will eventually be needed to generate ephemeris residuals for difficult targets.

Stacking

Normally, Tycho Tracker will stack your images as you expected when setting up the missions. That is to say that images from a mission (or pair of consecutive missions) with be stacked. However, if the groups do not contain the same number of images, Tycho Tracker may select the wrong grouping for the image stacks.

For example, 3 pairs of missions with 10 minutes between each pair, but some missions yielded 3 images and some only 2. Tycho Tracker may (for example) include one of the images from the last pair of missions in with the images from the second pair of missions. The time stamp for the stack will end up somewhere halfway between the second and third group of images, which is not what we want.

To solve this problem, after attaching ephemeris (as described above), in Image Manager select only the images to be included in the first stack then:

- Image Viewer/Create Stack Ephemeris.
- Double click to centre the target.
- Image Viewer/Create Observation.

Repeat this for the other groups/stacks of images.

When creating single observations like this, they will accumulate in the Observations-Single Target window. When you have collected all the observations select Add to Target List. You may need to insert the object's designation or number in the Object Designation window.

Quality and Residuals

Check residuals as you did in Level 1.

Up until now you have checked residuals using the on-line Find_Orb service. This is going to work for most targets. However, as mentioned above, you should consider the Find_Orb application as described in the Find_Orb Tutorial. This is quite a complex application but if you want to get seriously into Near Earth Object tracking then you should learn to use it.

What are reasonable residuals?

I would always consider residuals under 0.5" are "good enough" although I would prefer to get closer to 0.25".

For an object that has been recently observed by others, your residuals should be at least comparable to theirs. There is no point in submitting poor observations to the MPC if good observations are already in the database.

For an object that is in urgent need of observations, you can be more lenient.

Reports and Publications

Produce an ADES report

In the Observations – All Targets window, check your observations look good and check their residuals as you did in Level 1.

Then File/Save Observations to .obs file, so you can recall them later.

Report/Generate ADES report. In the ADES Report window, Copy to Clipboard then paste the report into an email and send it to your tutor with any relevant comments about the exercise.

Test the report

If necessary, Open Tycho and File/Load observations to retrieve the observations saved earlier.

In the Observations All Targets window Report/Generate ADES Report.

In the ADES Report window:

- Select Send to MPC.
- Set the ack email address as your own address, paul@slooh.com.
- Set the appropriate Target Type.
- Set the ack message to something meaningful such as Observatory code, date, name of target, type of object, your surname. E. g. “W88, 20250305, 2025 DU2, NEO, myname”.
- Set Mode to “Test” and Send.

The MPC should reply in the Send Report window within a few seconds. The response should end in “format valid”. If not, fix the problem and save the corrected report, or ask your tutor for help.

Send Report

When you have received the “format valid” message from the MPC, and have agreement from your tutor, change the Mode to “send” and select Send. You should receive a response within a few seconds. This will include an identifier which you should keep a note of.

Acknowledgement

If everything goes OK, you should receive an email acknowledgement within a few minutes. The acknowledgement only indicates that the report has been received and recognised as having observations in it.

The acknowledgement will include the contents of your acknowledgement message and some identifiers for each individual observation. Save this information. You can use these identifiers in the “Where are my observations” (WAMO) page.

Program Code

Each observer at each observatory has a “program code” for that observatory. It is recorded in the database to show who submitted what.

Your program code is assigned when you first submit observations from each observatory. It can take some time for the MPC to get round to assigning a new program code and during that period your observations will not be published. Check ”Where are my observations” to find out if your observations have been published.

Your program code for that observatory appears in the observation record as a one-character code that replaces the normal Note code.

```
K19O01U KC2019 08 22.91383 20 39 02.03 -21 43 58.1          16.8 GVEQ043L06
K19O01U KC2019 08 22.91733 20 39 01.61 -21 44 19.2          16.7 GVEQ043L06
K19O01U KC2019 08 22.92083 20 39 01.16 -21 44 40.6          16.9 GVEQ043L06
K19O01U 3C2019 08 23.01427 20 38 52.86 -21 53 42.5          16.3 GVEQ057G40
K19O01U 3C2019 08 23.01742 20 38 52.38 -21 54 02.0          16.4 GVEQ057G40
K19O01U 3C2019 08 23.02740 20 38 50.86 -21 55 03.5          16.3 GVEQ057G40
```

My program code at G40 is "3"

Publications

You will receive no feedback from the MPC regarding its acceptance or use of your observations but "no news is good news".

You may get an email if there is a problem with your report.

Once you have a program code, observations of NEOs will begin to appear in publications and databases within 24 hours of submission while observations of less critical objects may take a week or two.

Observation format

Observations published by the MPC are still in the old Obs80 (80 column punched card) format and they do not show all the details that were carried by the ADES report. (You can now retrieve ADES reports from the MPC but they are in JSON format and suitable only for computer processing).

NEOs and "Unusuals"

The easiest place to look for your observation is the observations database or the MPC Explorer.

NEO observations will appear quickly and there will be an MPEC (Minor Planet Electronic Circular) number against it. Links to recent MPECs can be found here.

Your observation will probably be in a Daily Orbit Update MPEC. You can check the Daily Orbit Update MPEC each day until you see your observations.

The MPEC will show each observation record slightly changed from your original submission. The "Note" code in each observation will be replaced by your Program Code and a special catalog code will be inserted in the space between magnitude and observatory code.

After a few days or weeks, the MPEC number against your observations in the database will change to a MPS (Minor Planet Circular Supplement) number. This can be retrieved from the archive and represents the permanent publication of the observations. MPSs are published about once per week but can get a bit irregular when the MPC is busy.

Ordinary Asteroids

Observations of Main Belt asteroids may take a few days or weeks to appear. This is because they are batched up and processed periodically, usually when other work is quiet around the full Moon. There will be no MPEC number, but the observation will eventually appear with an MPS number that can be retrieved from the archive.

Minor Planet Electronic Circular (MPEC)

MPECs are published whenever there is any worthwhile news. This can be many times every day. Each time a new object is designated or a "lost" object is recovered, it gets its own MPEC. Other NEO observations are batched up into a Daily Update MPEC. Some MPECs contain huge batches of orbit updates as the MPC works its way through calculating new orbits for all the known objects.

Minor Planet Circulars (MPC)

MPCs are published about once per month and (among other things) they summarise which objects have been observed by who at which observatory. Your name and Program Code will appear together with a list of objects you have reported on.

NEODyS and AstDyS

These databases will reflect your observations soon after the MPC has published them. They include residuals and indicators as to whether your observations were accepted for use in calculating the latest orbit and calculating the latest estimate of absolute magnitude (H).

Level 2 completed!

Congratulations on getting this far! It is worth practising what you have learned on a few different objects using different telescopes then move on to Level 3.

Tony Evans, last update Nov 2025